

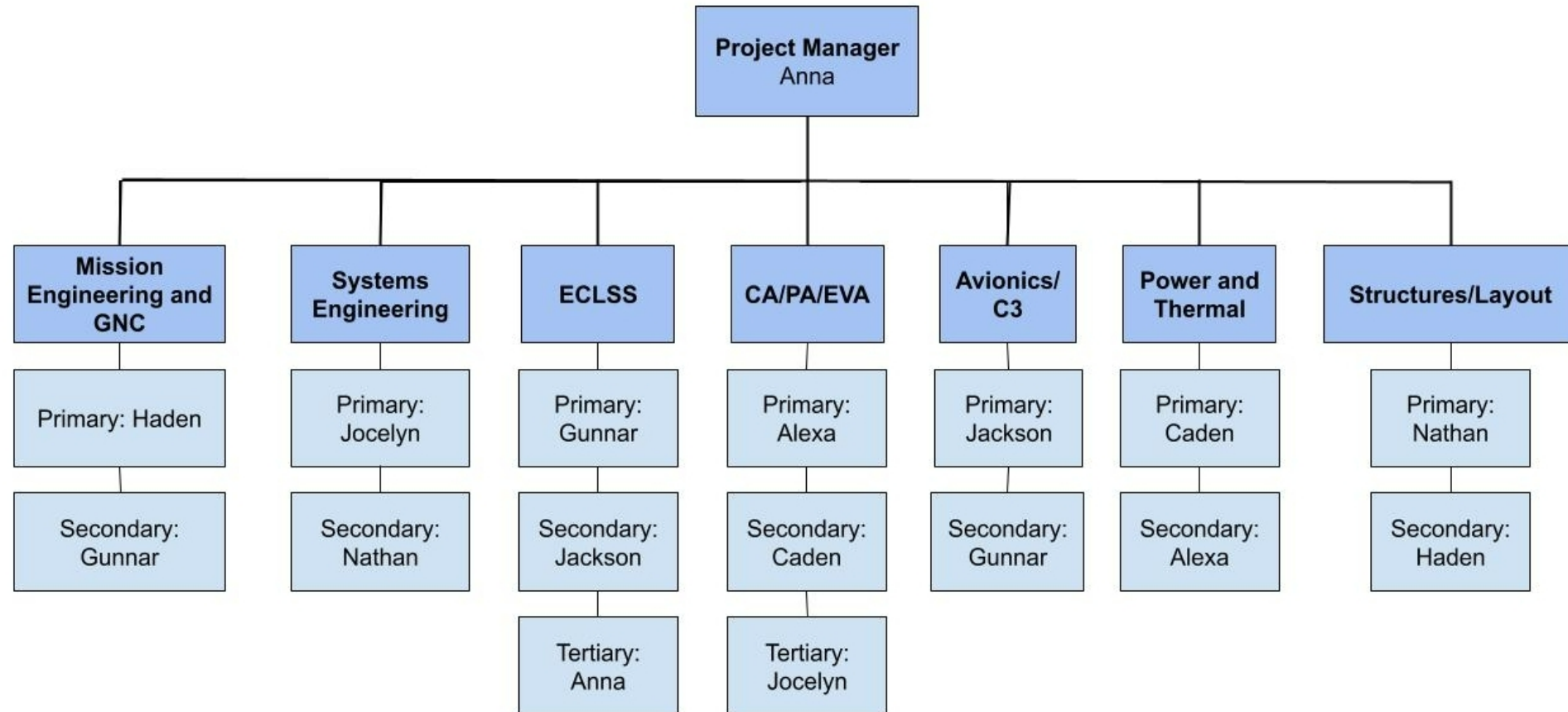
Team 2: Erebus Mission on Boyd Station

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500 km Habitat Design Project
University of Colorado Boulder

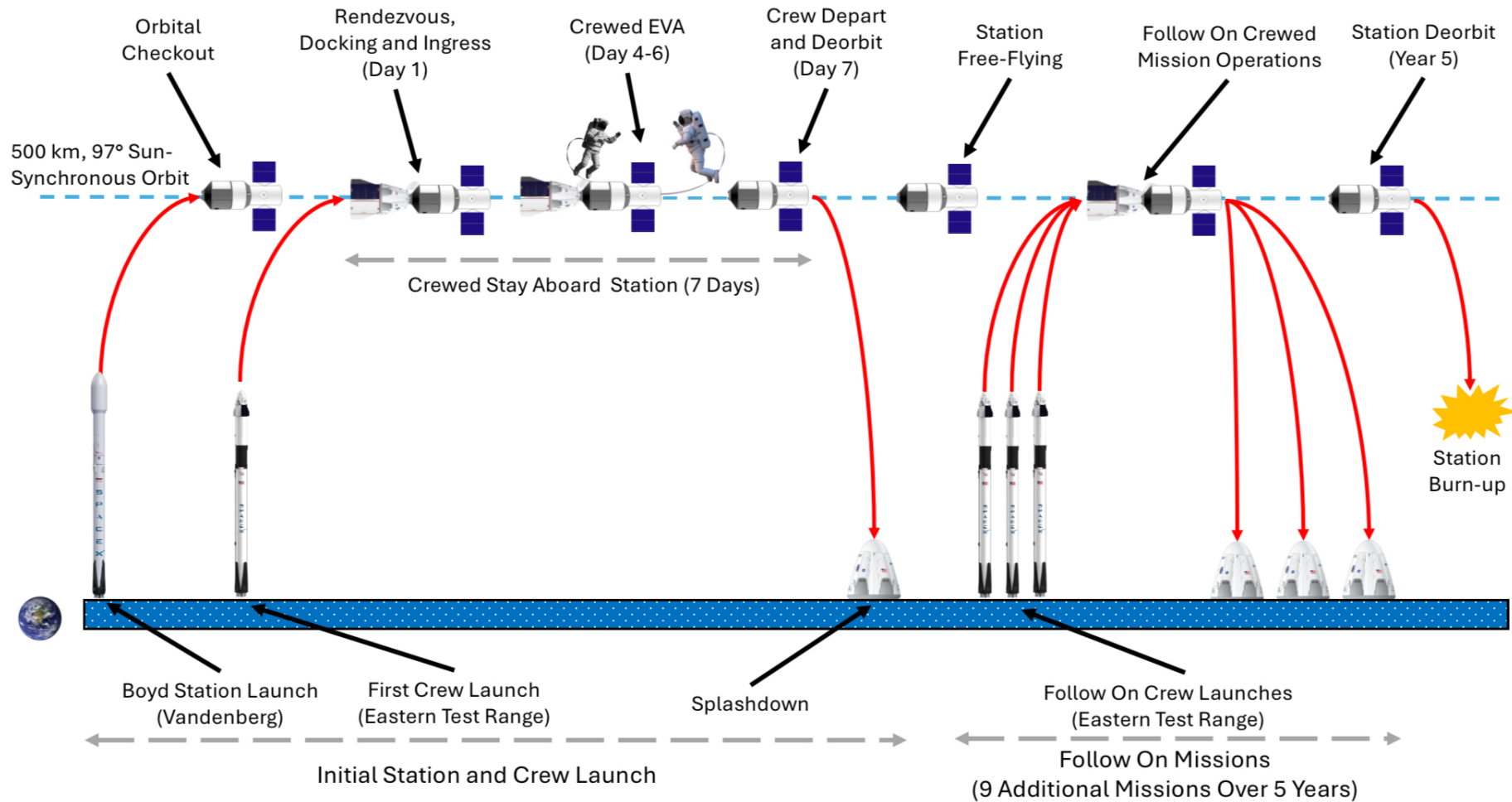


College of Engineering & Applied Science
UNIVERSITY OF COLORADO BOULDER

Team Organizational Chart



ConOps

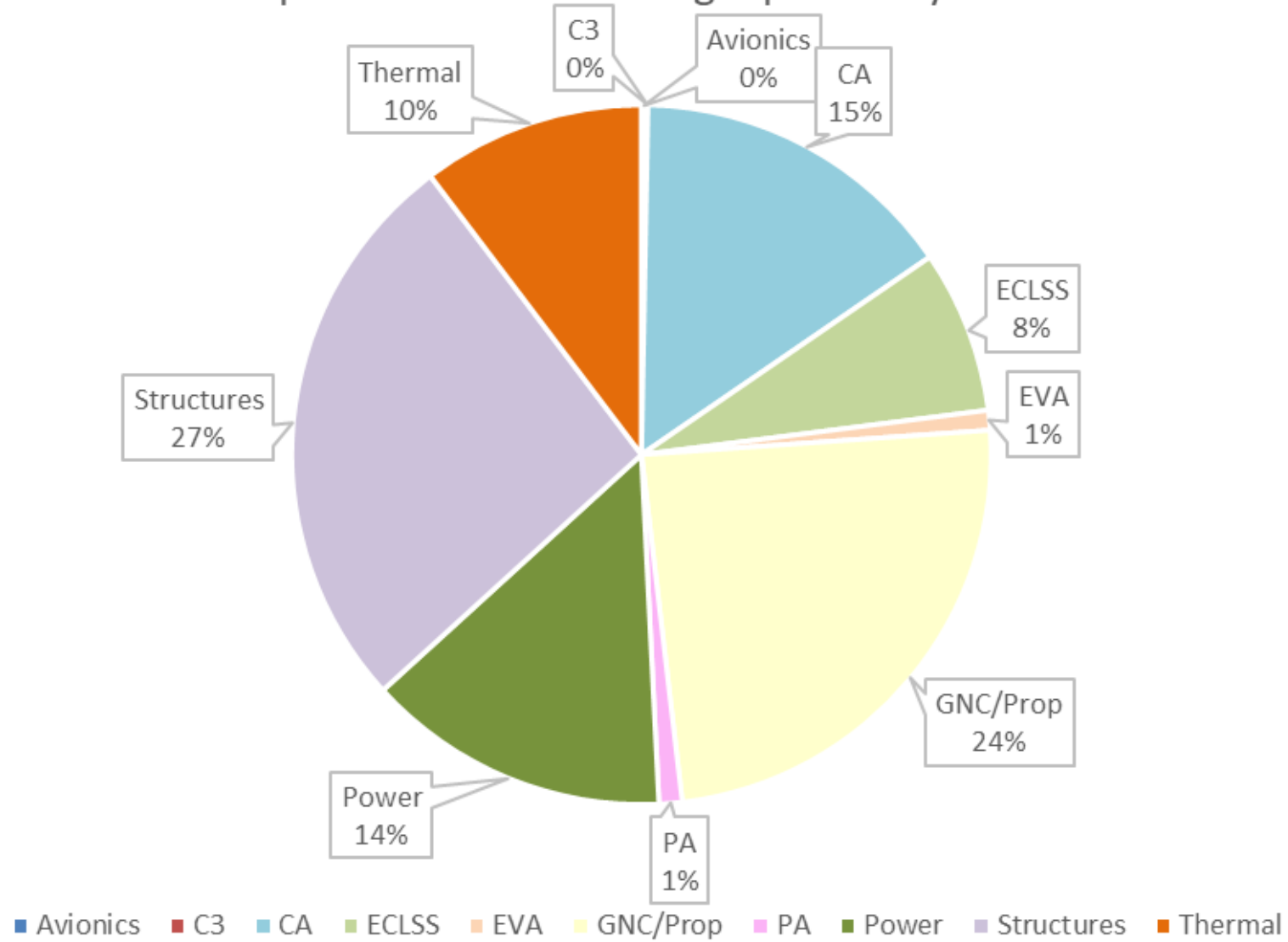


Mass, Power and Volume Totals

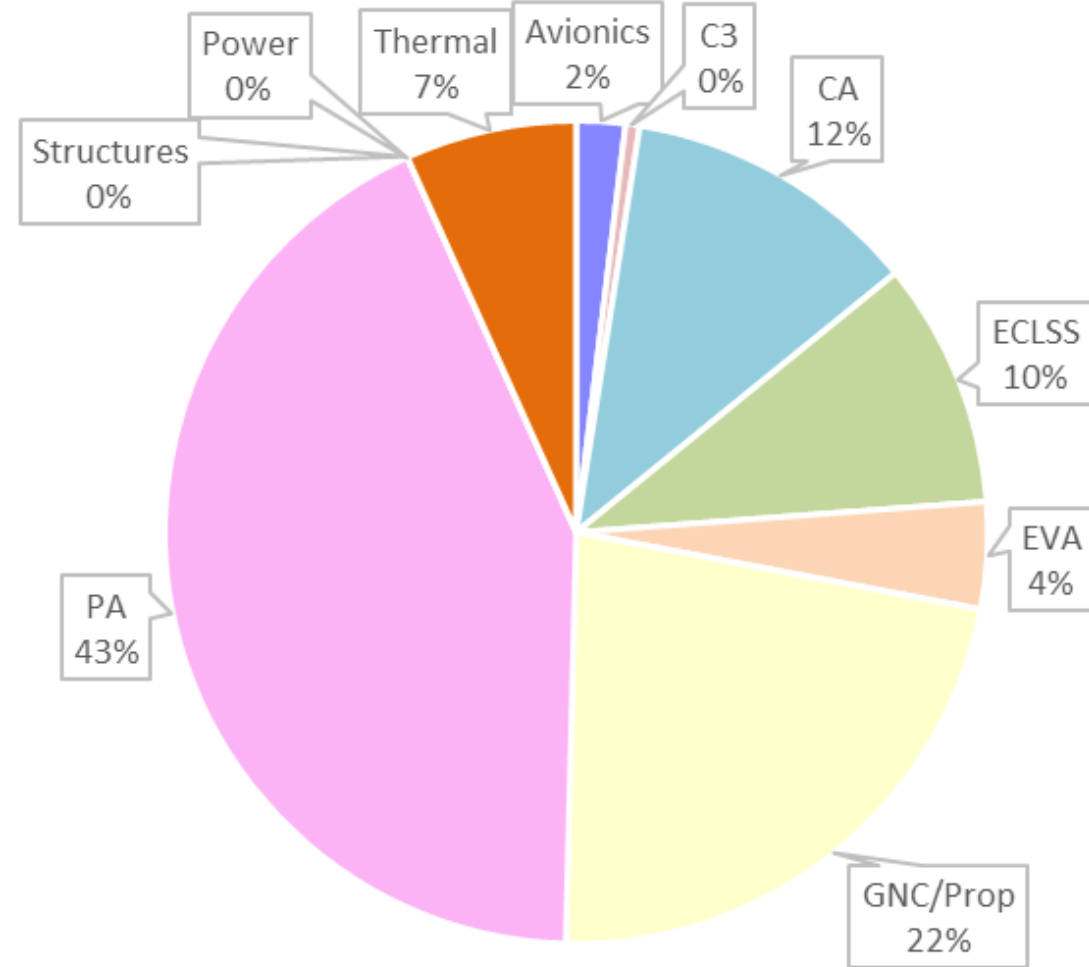
Subsystem	Bottom-up w/ margin (kg)	Total Power (W)	Total Volume (m ³)
Avionics	27	47	0.028
C3	8	14	0.011
CA	1822	286	11.676
ECLSS	906	237	5.614
EVA	110	103	0.282
GNC/Prop	2920	549	1.140
PA	127	1050	0.277
Power	1690	0	3.534
Structures	3181	0	4.881
Thermal	1242	165	7.570
Total:	12037	2450	35.013



Bottom-up Wet Mass with Margin per Subsystem



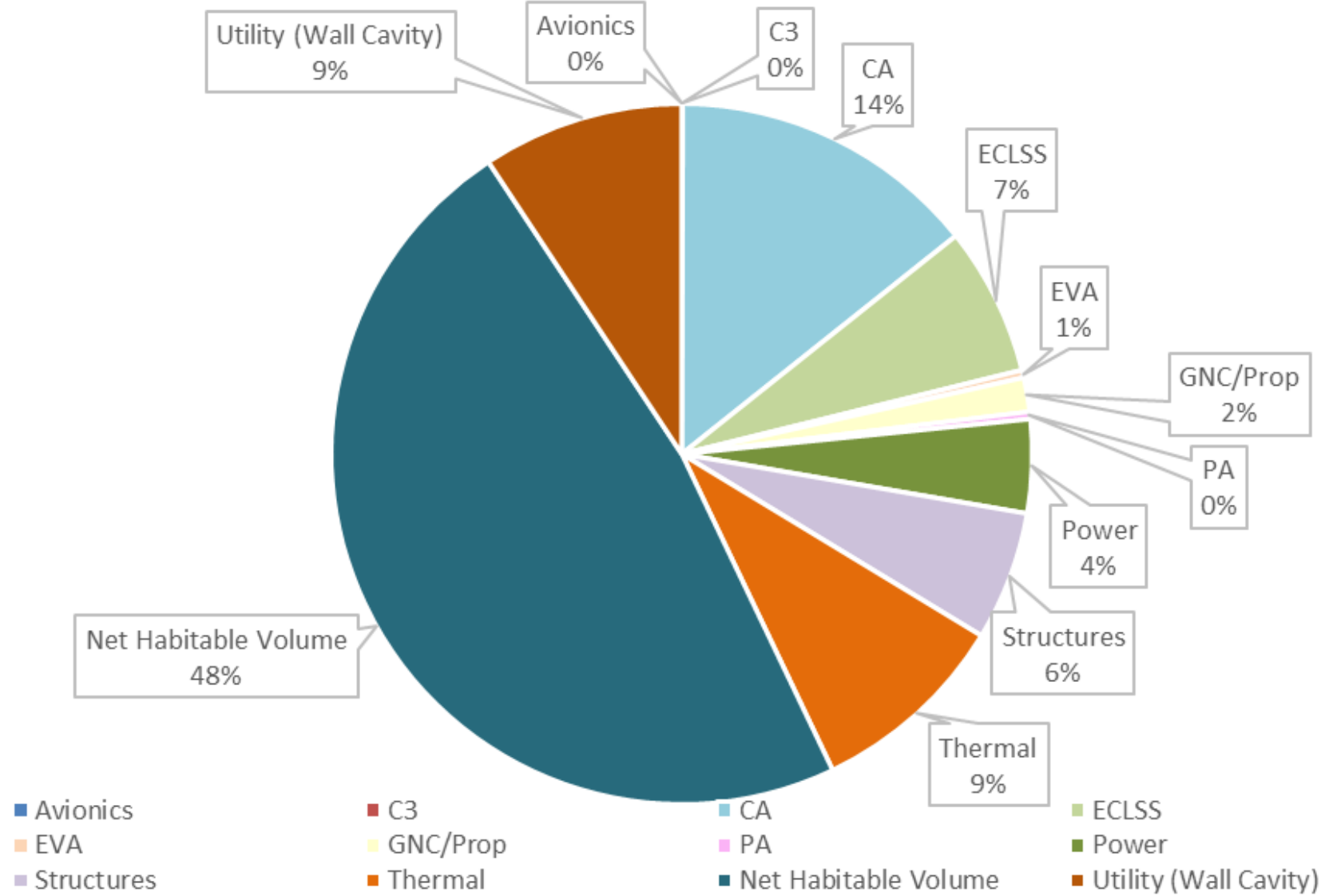
Power Demands per Subsystem



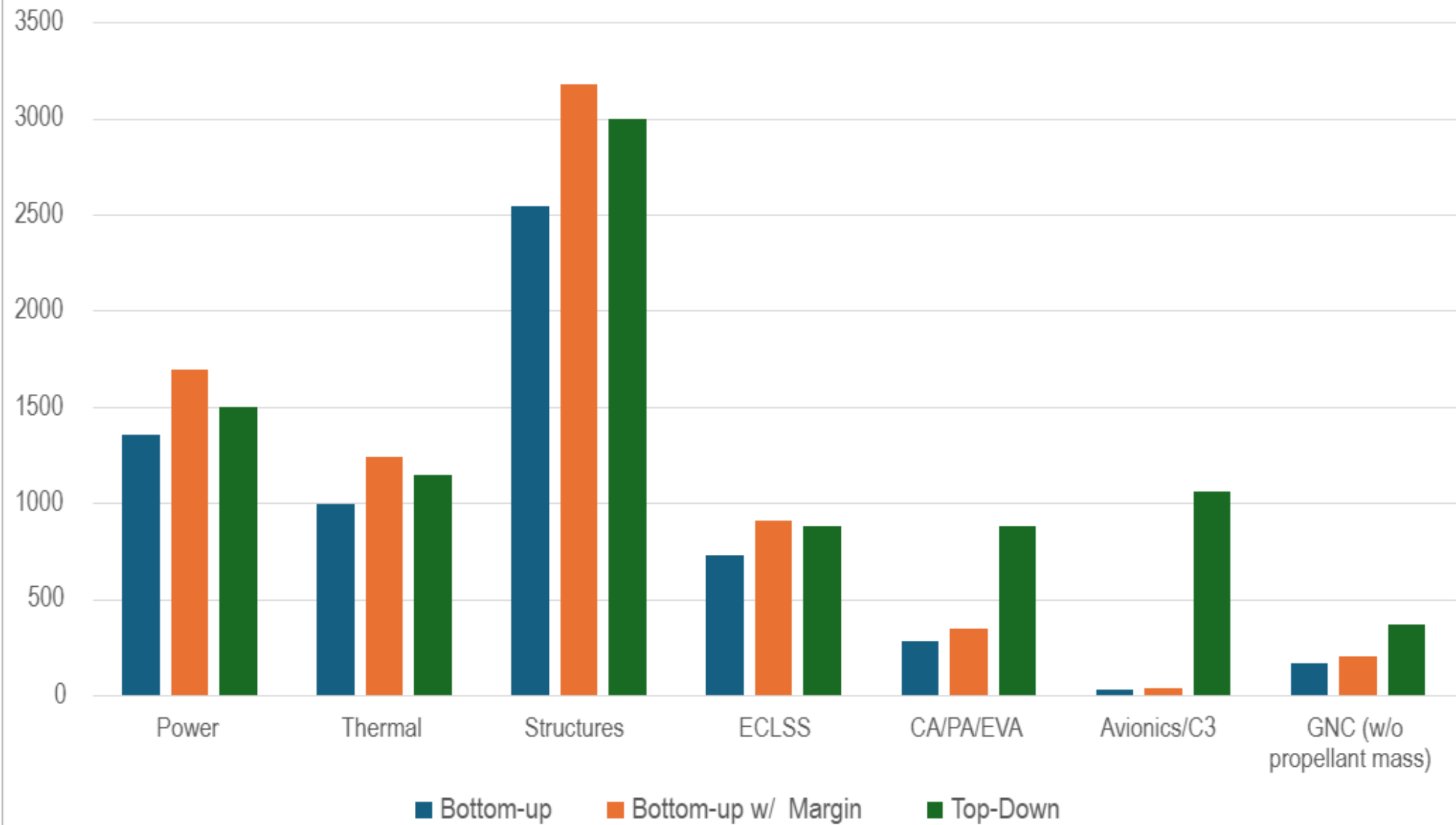
Avionics C3 CA ECLSS EVA GNC/Prop PA Power Structures Thermal



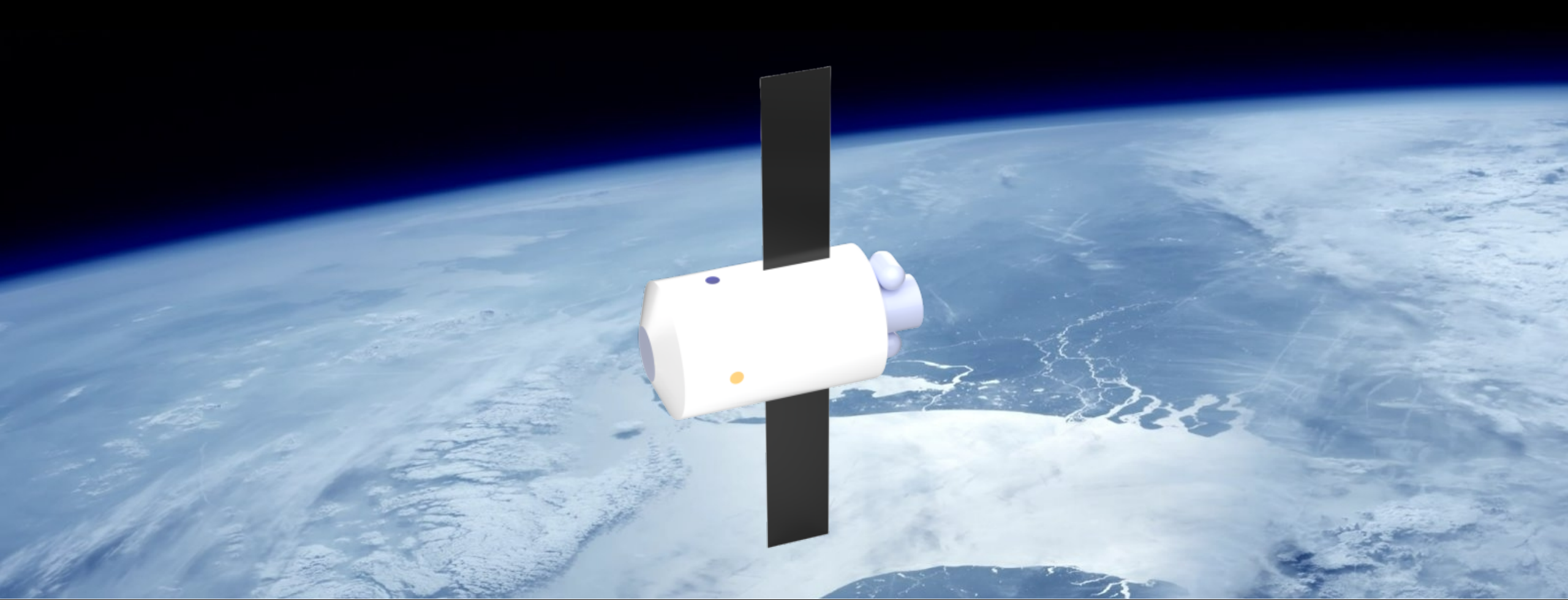
Total Volume per Subsystem



Mass Analysis Comparison by Subsystem



Structures/Layout

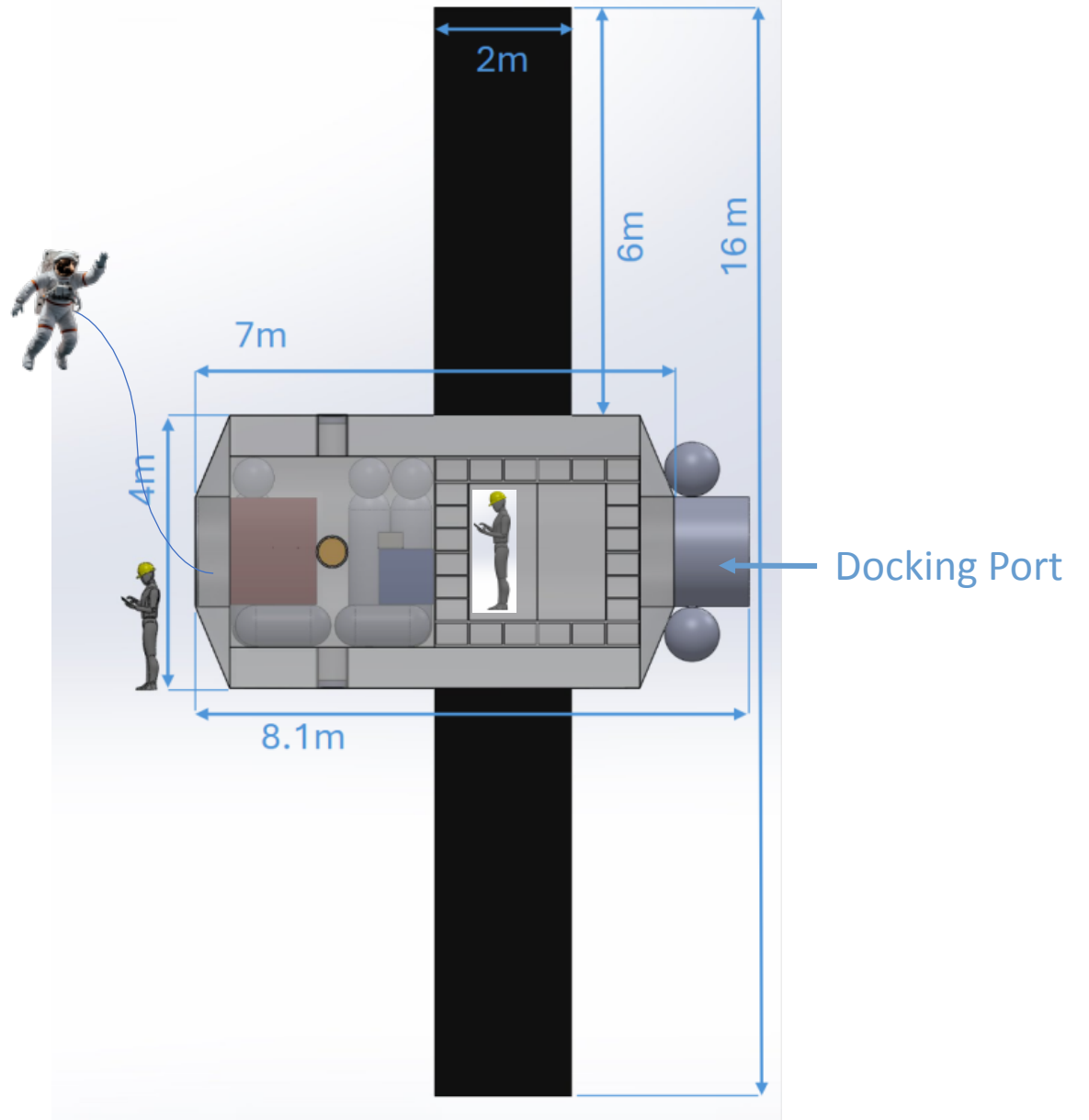


Structures/Layout

Total Pressurized Volume: $\sim 82\text{m}^3$

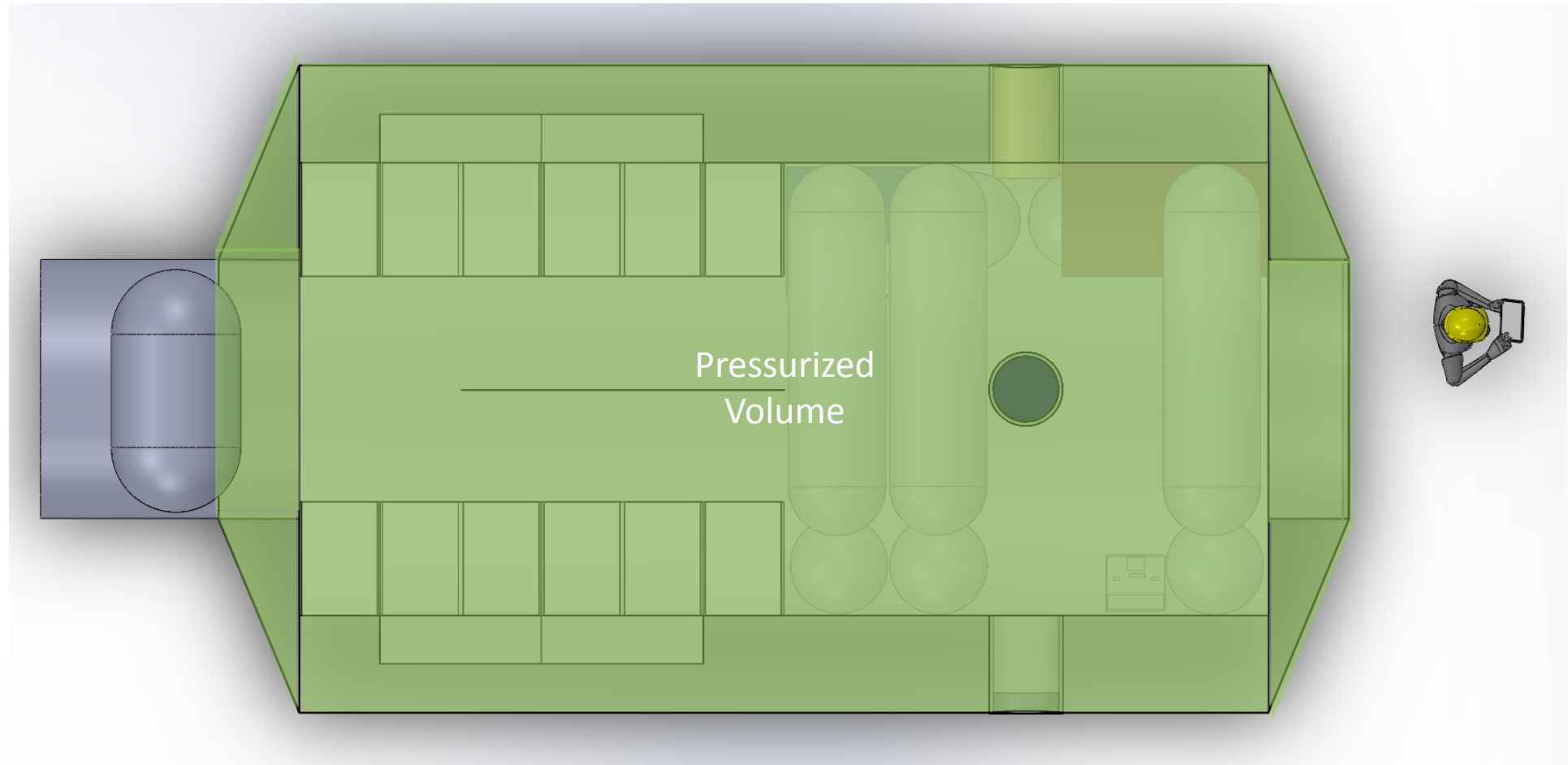
Habitable Volume: $\sim 39\text{m}^3$

**For storage considerations the EVA Suits will be stored in the crew transport vehicle pre and post-EVA*

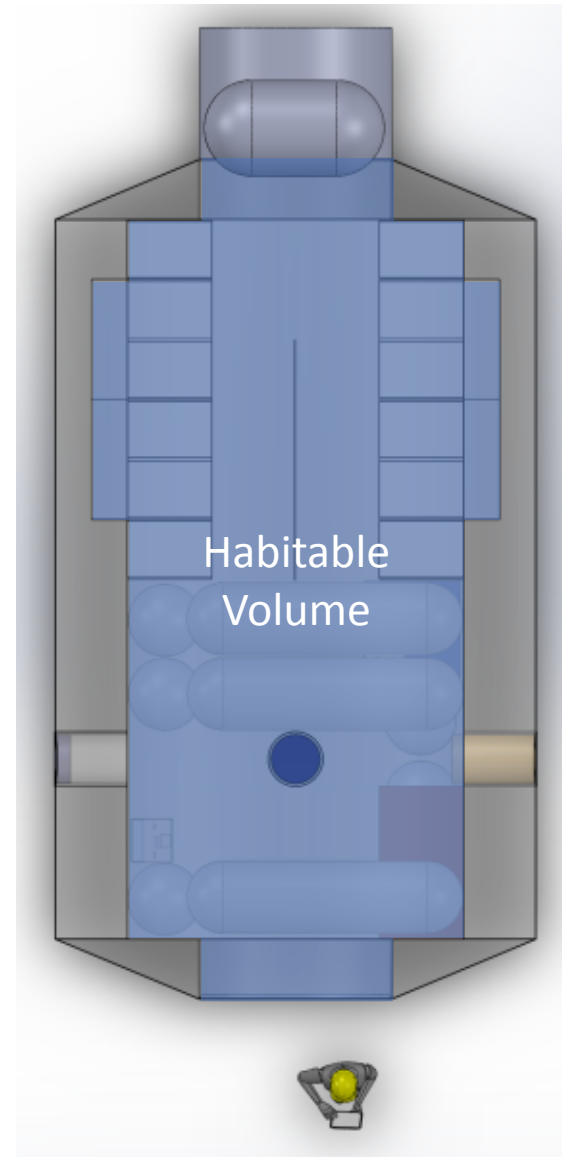
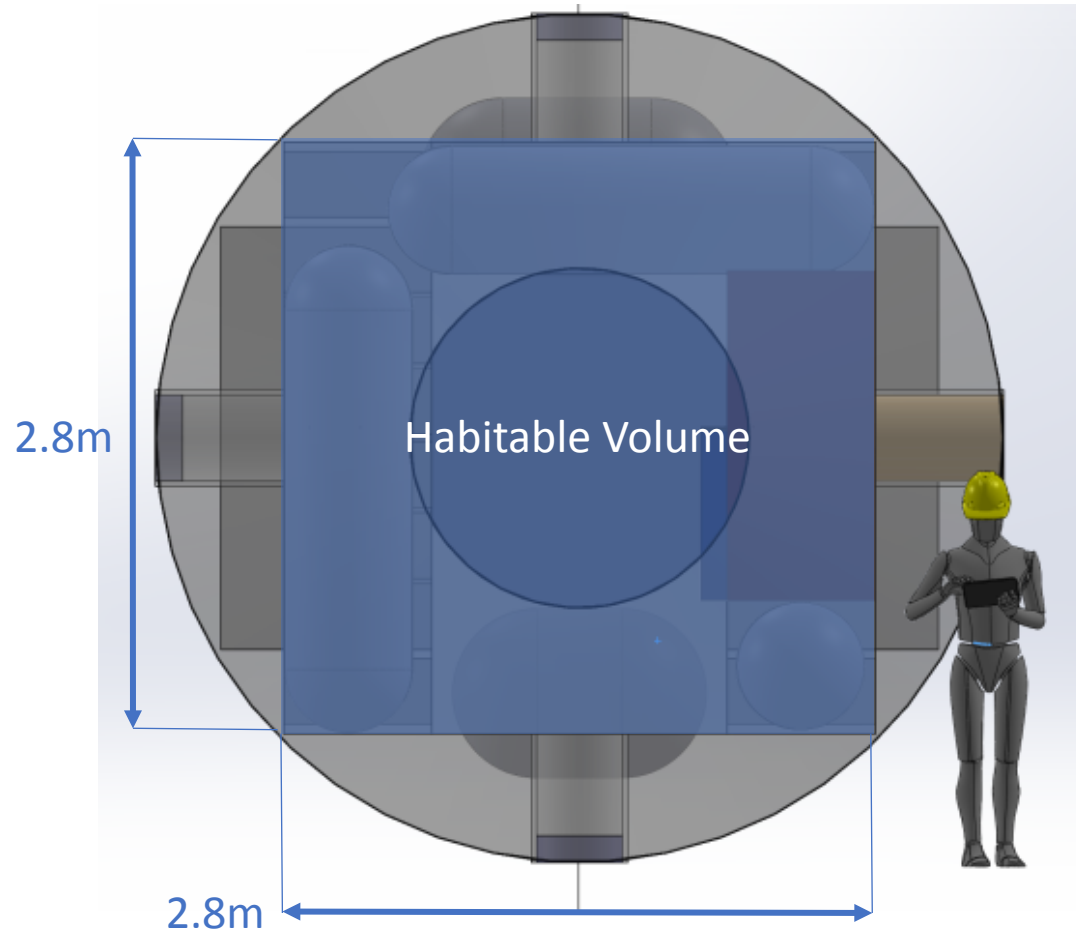


Structures/Layout

Total Pressurized Volume: $\sim 82\text{m}^3$



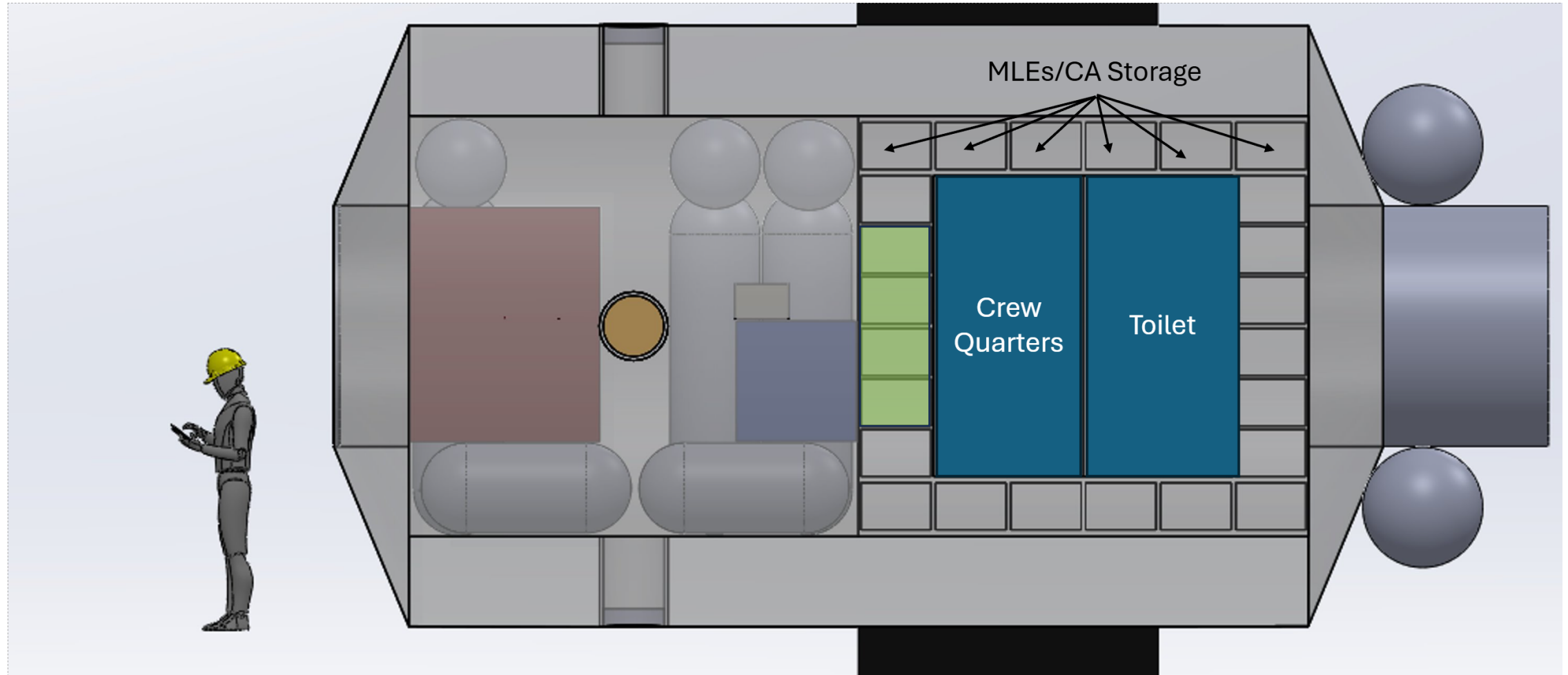
Structures/Layout



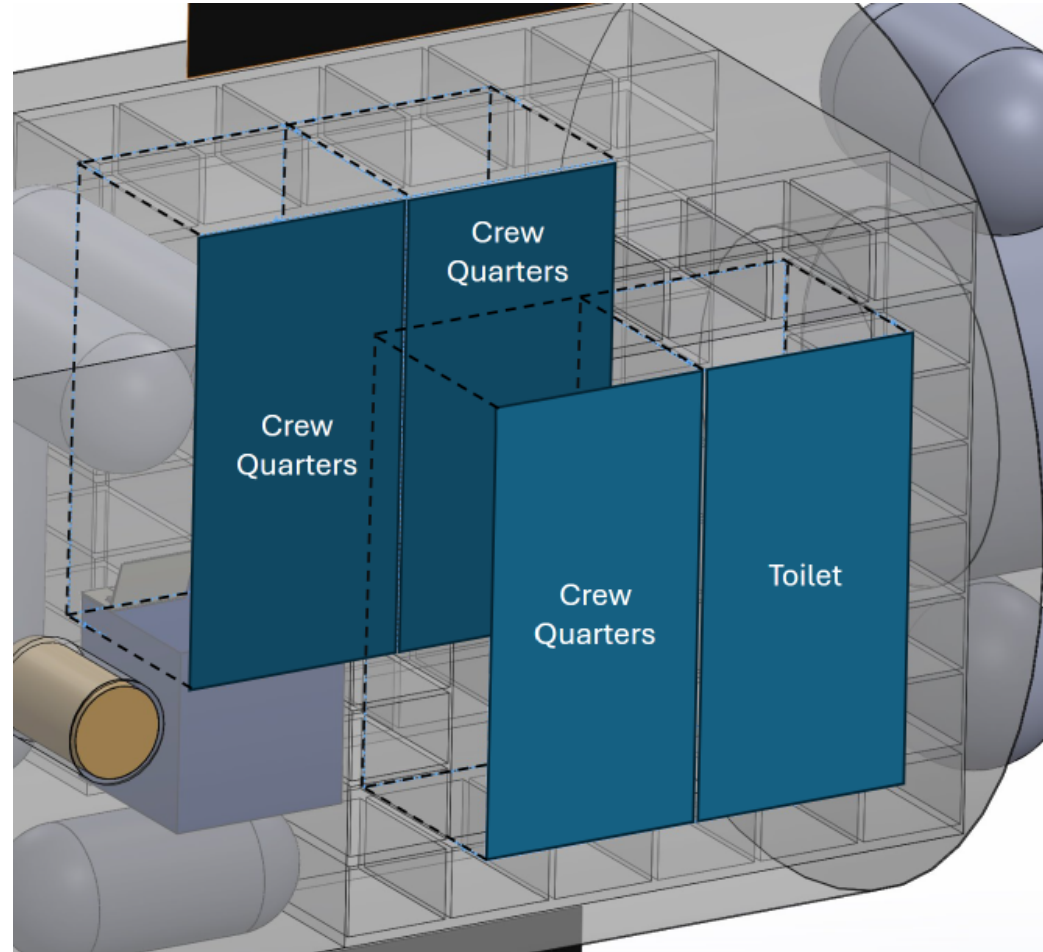
Habitable Volume: $\sim 39\text{m}^3$



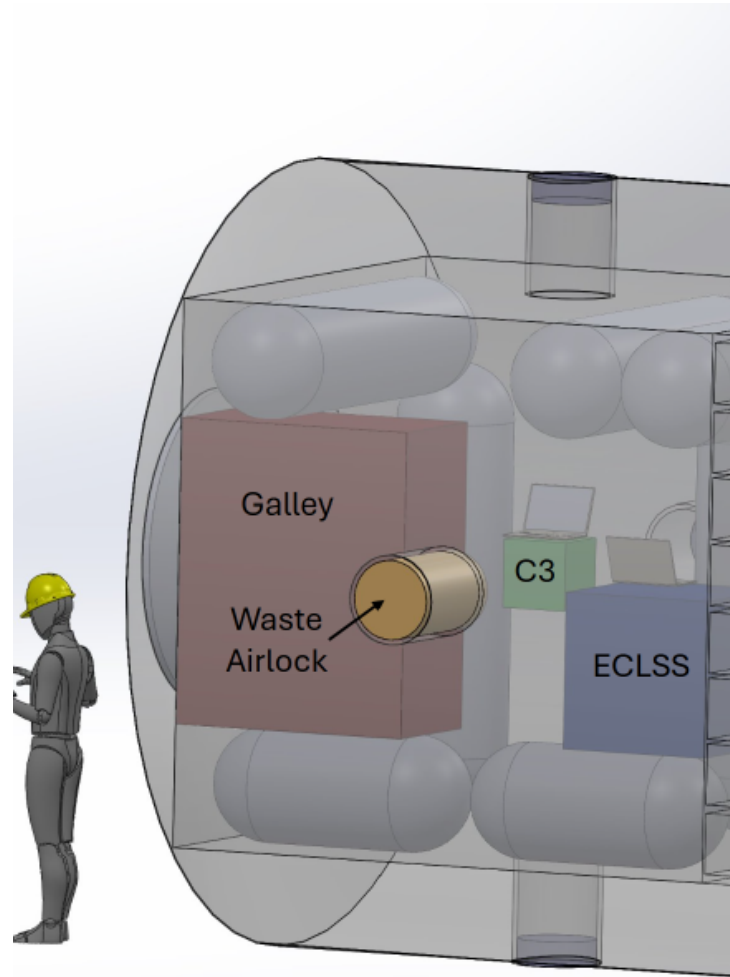
Structures/Layout



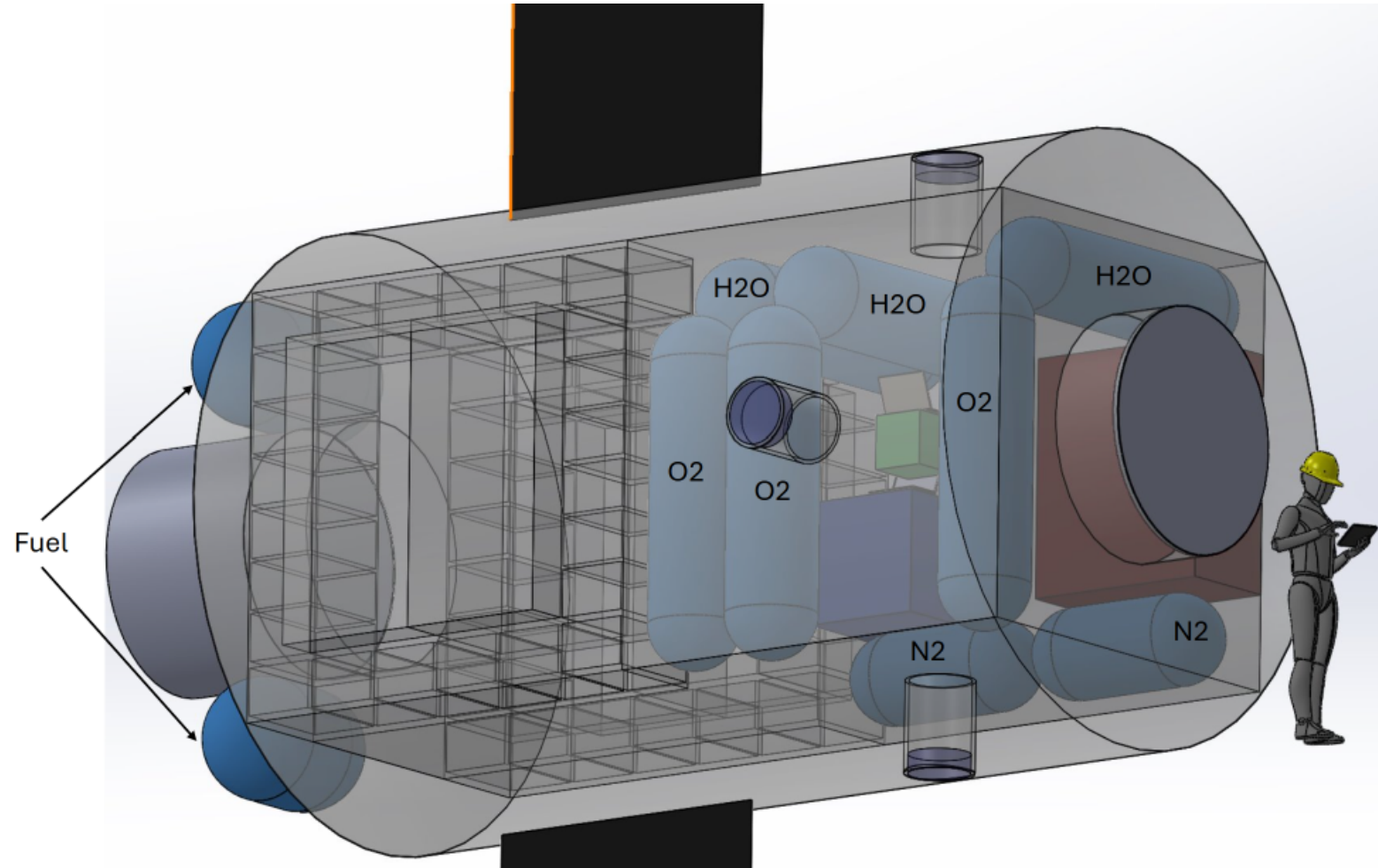
Structures/Layout



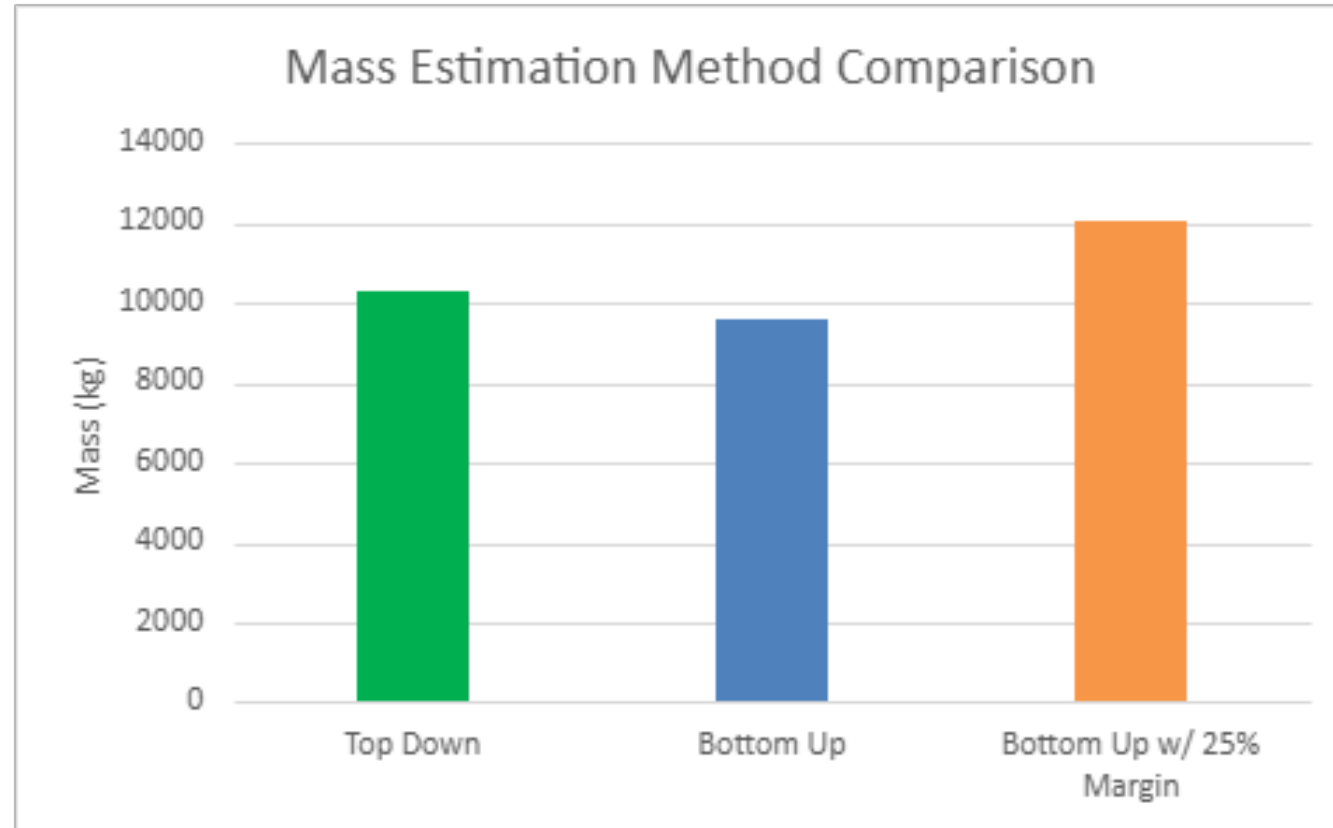
Structures/Layout



Structures/Layout



Overall integrated total Launch mass



Normalization Ratios – Volumetric Efficiency

$$\text{Volumetric Efficiency} = \frac{\text{Net Habitable Volume}}{\text{Total Pressurized Volume}} \cdot 100$$

Net Habitable Volume = 39.150 m³

Total Pressurized Volume = 81.932 m³

Volumetric Efficiency = 47.78%



Normalization Ratios — Packaging Density

$$\text{Packaging Density} = \frac{\text{Total Habitat Mass (with margin)}}{\text{Total Pressurized Volume}}$$

Total Habitat Mass = 12037 kg
Bottom-up wet mass w/margin

Total Pressurized Volume = 81.932 m³

$$\text{Packaging Density} = 147 \text{ kg/m}^3$$



Normalization Ratios – Consumable Mass Fraction

(i) $Consumables = C_{ECLSS} + C_{CA} + C_{PA} + C_{EVA} + Fuel = 2893 \text{ kg}$

(ii) $Total \text{ Dry Mass} = Total \text{ Wet Mass (with margin)} - Consumables = 9145 \text{ kg}$

Total Wet Mass = 12037 kg is the bottom-up wet mass with 25% margin

(iii) $Consumable \text{ Mass Fraction} = \frac{Total \text{ Dry Mass}}{Total \text{ Wet Mass (with margin)}} \cdot 100$

$Consumable \text{ Mass Fraction} = 75.97\%$

CONSUMABLES	
Subsystem	Total Mass (kg)
ECLSS	462
CA	112
PA	100
EVA	41
Fuel	2178
Total	2893



Pressurized Volume Per Crewmember

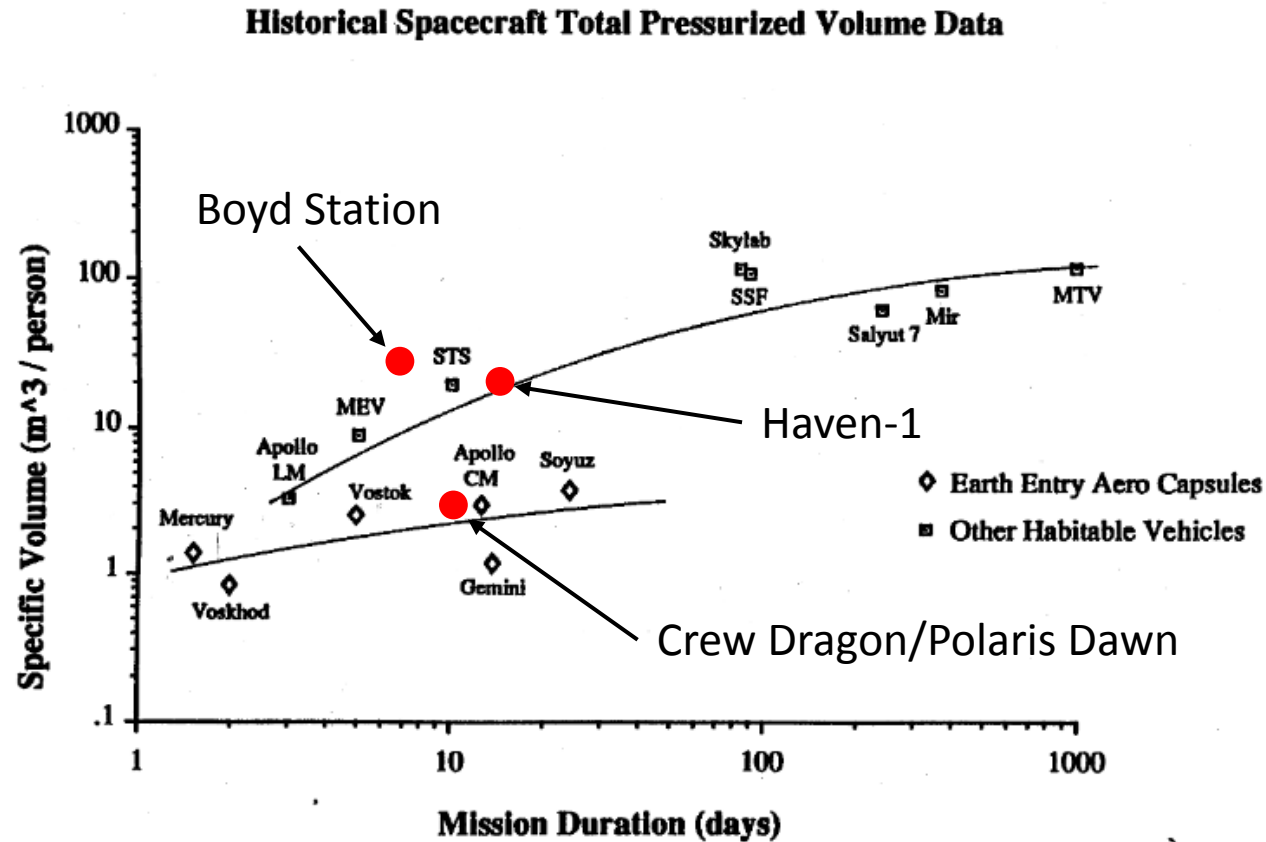
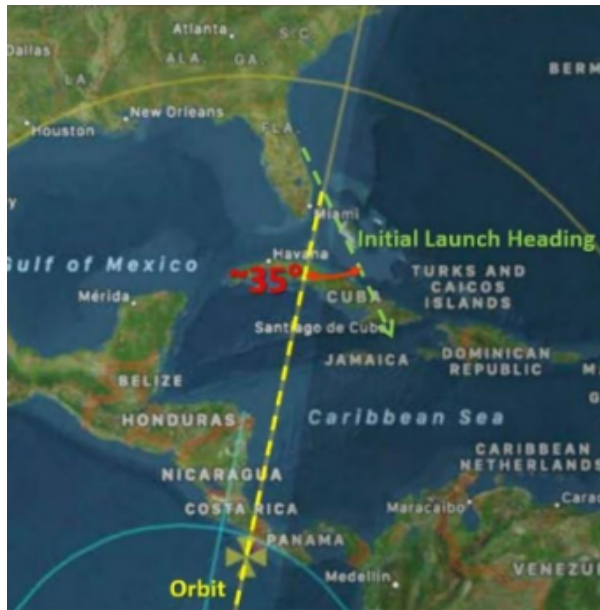


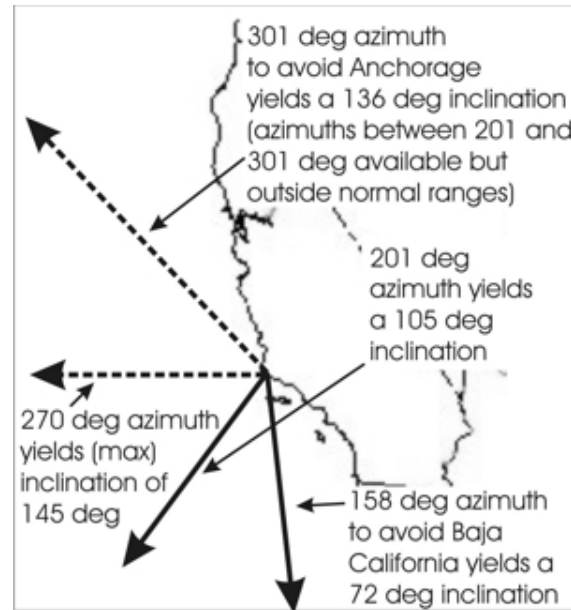
FIGURE 9. Sherwood and Capps, 1990, separation of two curves that distinguish aero-entry capsules and all other "habitable vehicles" (courtesy of Brent Sherwood).



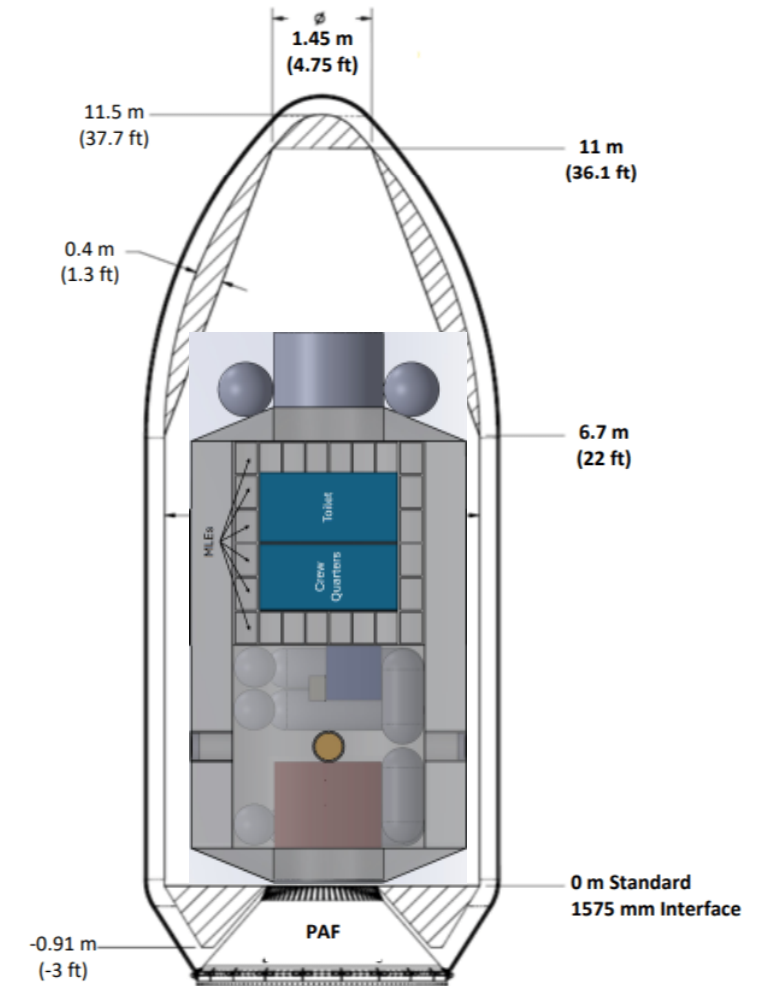
Launch Vehicle selection



Falcon 9 Crew Dragon from Kennedy Space Center



Falcon 9 Full Thrust from Vandenberg Space Force Base



Comparison to other Commercial Habitats

***Volumetric Efficiency* = 47.78%**

Boyd Station

VAST's Haven-1:

Habitable Volume = 45 m³

Pressurized Volume = 80 m³

Volumetric efficiency = 56.25%

***Packaging Density* = 147 kg/m³**

Boyd Station

VAST's Haven-1:

Mass = 14600 kg

Packaging density = 182 kg/m³



Key Take Aways

- Scope creep could have easily influenced the project's development.
- A focused design early can prevent scope creep when evaluating new possibilities.
- Human life support systems are less demanding on mass, power, and volume than anticipated.

